Numerical Validation of the Shear Compression Specimen (SCS). Part I: Quasi-Static Large Strain Testing

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ABSTRACT

The shear compression specimen (SCS), which is used for large strain testing[7], is thoroughly investigated numerically using three-dimensional elastoplastic finite element simulations. The first part of this study addresses quasi-static loading. A bi-linear material model is assumed. The effect of geometrical parameters, such as gauge height and root radius, on the stress and strain distribution and concentration is investigated. The analyses show that the stresses and strains are reasonably uniform on a typical gauge mid–section, and their average values reflect accurately the prescribed material model. Accurate correlations between the averaged von Mises stress and strain and the applied experimental load and displacement are derived. These relations depend on the specimen geometry and the material properties. Numerical results are compared to experimental data, and an excellent agreement is observed. This study confirms the potential of the SCS for large strain testing of material.

KEY WORDS- large strain, shear compression specimen, elastic plastic material, finite elements, quasi-static

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Introduction

Large shear strains are found in many applications, such as (high speed) machining, ballistic failure, ductile fracture, and micromechanics of void growth and coalescence. The large strain constitutive behavior of a material has therefore to be determined for modeling and calculation purposes. However, inherent experimental limitations such as necking, barreling or buckling, all affect the various specimens to date, so that large strains are not easily obtainable. Numerous examples can be found in the literature, of original specimen analyses and/or testing procedures used to investigate the large strain plastic properties of materials, such as sheet metal[1] and composites under shear[2]. Other examples can be found in the work of Khan and Wang[3] who developed a two stage procedure to achieve finite strains in 1100 aluminum, with emphasis on modeling various loading paths. Wang and Lee[4] proposed a numerical modeling procedure to address wrinkling problems in thin sheets. Nouailhas and Cailletaud[5] used crystal plasticity in conjunction with finite element simulations to investigate single crystal superalloys. Moreover, when shear loading is addressed, the load-displacement curves are readily translated into shear stress-shear strain curves, assuming that shear is the only mode of deformation. However, the use of various specimens, including even the "simplest" torsion cylinder, or more sophisticated shear specimens[2], may require numerical simulations to assess the validity of certain assumptions, such as homogeneity of the stresses and strains. Numerical simulations are also frequently used to develop various dynamic tests, such as adiabatic shear testing[6]. It also seems that, in most cases, the experimentalist must use several specimen geometries to cover a wide range of strain rates and large strains if possible. Consequently, one would want to use a single specimen geometry to cover in a *seamless* manner the largest possible range of strain rates while allowing for large strains to develop. The shear compression specimen (SCS), suggested by Rittel et al.[7,8] , was specially designed to answer these needs. The specimen, of a very simple geometric shape, consists basically of a cylinder or a parallelepiped with a couple of opposite slots machined at 45° on its faces. As the specimen is axially loaded (either dynamically or statically), the gauge (slot) experiences a dominant state of shear. However, as emphasized in all the above-mentioned references, one should not assume a state of pure shear, as preliminary numerical calculations[7] clearly showed that the state of stress is three-dimensional. Consequently, it had been proposed to base the analysis on the Von Mises equivalent stress and strain definitions. The validity of the specimen was further established by comparing stress-strain data obtained with the SCS to those obtained from uniaxial (tension/compression) tests[7-9]. However, while the first report[7]

suggested simple "universal" approximations for the effective stress and strain as a function of the load and displacement respectively, further work suggested that these approximations should be fine-tuned to the material itself and to the gauge geometry[9]. Several fundamental issues were only addressed in a preliminary manner to date, in the sense that the numerical results were based on a *single* representative finite element at the center of the gauge section, without further averaging of the mechanical fields. Issues, such as the uniformity of the stress and strain throughout the gauge, as well as the nature of the stress concentration in the gauge fillet and its influence on the determination of the mechanical properties, still remain open. Finally, the central assumption that the same approach (data reduction technique) applies identically to quasi-static *and* dynamic testing was only validated experimentally to date[9]. Consequently, a comprehensive numerical study was carried out to assess the advantages and limitations of this new specimen, with emphasis on the above-mentioned issues and keeping in mind simple stress-strain data reduction techniques. The two-part study addresses both quasi-static and dynamic loading conditions. The first part concerns quasi-static loading.

The paper is organized in four sections. The first section introduces the subject of this work. The second section presents the investigation of the SCS subjected to quasi-static loading. The results are discussed in the third section, followed by a concluding section.

The second section is divided into seven subsections. The first subsection introduces the SCS geometry and the material properties which were used in this study. The numerical model and data processing technique are detailed in the second subsection. The third subsection presents the numerical results. These results address issues that were not previously addressed, such as: 1. The effects of different plastic moduli, gauge height, and yield stress on the load-displacement curves. 2. The evolution and uniformity of the plastic zone. 3. The influence of the plastic modulus and gauge height on the averaged values of stresses and strains in a typical mid-section of the gauge. 4. The influence of gauge root radius. The fourth subsection introduces mathematical approximations of averaged stresses and strains. This subsection describes the relationship between the averaged von Mises stress and strain to the applied load and displacement respectively. A general approach for the assessment of the plastic modulus is developed in the fifth subsection. Two experimental procedures for the determination of the plastic modulus in a quasi-static test are introduced. These procedures are followed by a reintroduction of a simple stress-strain determination procedure of highly practical relevance to the experimentalist. The dependence of its three coefficients on the plastic modulus and gauge width is also detailed. Numerical verification problems are presented in sixth subsection, followed by a limited experimental validation.

Specimen Geometry and Material Data

The shear compression specimen (SCS) is shown in Fig. 1. As mentioned in the introductory section, this specimen promotes shear deformation in an inclined gauge-section, and it is aimed at large-strain constitutive testing of materials under both quasi-static and dynamic loading conditions[7, 8] For all the investigated cases, the length, diameter, gauge thickness and root corner radius were chosen to be: $L = 20$ mm, $D = 10$ mm $t = 2.5$ mm and $R_1 = 0.125$ mm respectively. Four different slot widths were selected $w = 0.25$ mm, $w = 0.5$ mm, $w = 1.0$ mm and $w = 2.0$ mm. The influence of three different root corner radii $R_1 = 0.125$ mm, $R_1 = 0.25$ mm and $R_1 = 0.4$ mm was also investigated for a gauge width of $w = 1.0$ mm. These sets of geometrical parameters cover most of the actual experimental conditions.

The specimen material is assumed to be bi-linear with Young's modulus $E = 210$ GPa, Poisson's ratio $\nu = 0.3$ and yield stress $\sigma_Y = 0.7$ GPa. Three representative hardening moduli were analyzed: $E_p = E/100$, $E_p = E/500$ and $E_p = E/2000$. These values are meant to simulate materials with "high", "average" and "low" hardening characteristics, again to represent a vast class of material behaviors.

The Numerical Model

The numerical analysis was carried out using the commercial finite element code ANSYS[10]. Due to the symmetry of the problem, only one half of the specimen is modeled as can be seen in Fig. 2. A typical mesh, which consist of 20335 elements of ANSYS type SOLID187 (3-D 10-Node Tetrahedral Structural Solid), is shown in Fig. 2.

Traction free boundary conditions were applied on all surfaces, except for surfaces A9, A11 and A13 as shown in Fig. 3. Symmetry conditions were enforced on surface A13, while surface A9 was constrained from vertical (y) motion and totally constrained at an arbitrary point. A displacement boundary condition of $d = -1.75$ mm along the y-direction was applied on surface A11.

The nonlinear equilibrium equations are solved implicitly and incrementally (see ANSYS[10] ,Theoretical Manual). The non-linearity includes material and geometrical effects, namely plasticity and large strains. At each load step i $(1 < i < N)$ the applied displacement d_i is known, and the resulting pressure force P_i at the upper surface (A11) is calculated by integrating the normal stresses σ_{yy} over the surface A11.

Averaged values of the equivalent von Mises stress $(\hat{\sigma}_{eqv})$, hydrostatic stress $(\hat{\sigma}_{hyd})$, and the equivalent total strain $(\hat{\varepsilon}_{eqv})$ on the inclined mid-section of the gauge are calculated at each load step. The inclined mid-section of the gauge is shown in Fig. 4, where the upper half of the specimen has been removed. The averaged values are calculated by integrating the stresses on the mid-section and dividing them by the area of the plane, according to:

$$\hat{\sigma}_{eqv} = \frac{1}{A}\int_A \sigma_{eqv}\, dA \qquad (1)$$

$$\hat{\varepsilon}_{eqv} = \frac{1}{A}\int_A \varepsilon_{eqv}\, dA \qquad (2)$$

$$\hat{\sigma}_{hyd} = \frac{1}{A}\int_A \sigma_{hyd}\, dA \qquad (3)$$

Results

Load-Displacement

During an experiment, the enforced displacement (d) on the upper surface (A11 in Fig. 3) and the resulting (negative) load (P) are measured. The resultant P-d curves for four different gauge heights $w = 0.25, 0.5, 1.0, 2.0$ mm are plotted in Fig. 5. Two hardening moduli were considered: $E_p = E/2000$ ("soft") and $E_p = E/100$ ("hard"). All the curves coincide until the load reaches -18000 N (approximately), as a macroscopic indication for the end of the elastic region. This figure shows that for a decreasing gauge height, a higher compressive load is needed to enforce the same displacement. The bi-linear nature of the

constitutive equations is reflected on the P-d curves for the "soft" material while the "hard" material exhibit a non-linear behavior, as noticeable from the curvature of the P-d curves.

The specimen with $w = 2\ mm$ was analyzed one more time, assuming now a yield stress $\sigma_Y = 1200$ MPa instead of $\sigma_Y = 700$ MPa. The resulting P-d plots are shown in Fig. 6. The analyses demonstrate that the yield load is proportional to the assumed yield stress, as expected.

Stress and Strain Distribution in the Gauge

This section first describes the stress and strain distribution in the whole specimen, followed by their distribution on a representative mid-section of the gauge.

A contour plot of the von Mises equivalent stress on surface A13 of Fig. 3 is plotted in Fig. 7 for two different hardening materials. The plot represents a specimen of w = 2 mm with prescribed displacement of d = 1.75 mm.

The minimum value for contour plot was chosen to be $\sigma_{eqv} = \sigma_Y = 700$ MPa. This value was chosen in order to emphasize the regions which have undergone plastic deformation. The plastic zones corresponding to $E_p = E/2000$ and to $E_p = E/100$ are shown in Fig. 7a, and Fig. 7b respectively. The plastic zone for $E_p = E/2000$ is mostly confined to the gauge, while for $E_p = E/100$ the plastic zone spreads into the whole specimen. This fact might explain the nonlinear behavior of the P-d curves for $E_p = E/100$ in Fig. 5.

An important point is that of the uniformity of stress and strain on a typical mid-section of the gauge. As in any experimental specimen geometry, one cannot expect these quantities to be strictly uniform, and this point is quite important to assess the quality of experimental results. We selected a specimen of 2 mm gauge height, subjected to a displacement of 1.75 mm, with $E_p = E/100$. This high plastic modulus value was selected to represent the "worst" case in which plasticity spreads out of the gauge. The distribution of σ_{eqv}, ε_{eqv} and σ_{hyd} on the mid-section is presented in Fig. 8.

Averaging σ_{eqv} by means of Eqs. (1) yields $\hat{\sigma}_{eqv} = 1.501$ GPa. Fig. 8a reveals that σ_{eqv} is not strictly uniform on the mid-section, but ranges between $0.87 \le \sigma_{eqv} / \hat{\sigma}_{eqv} \le 1.1$. Most of the area of the section experiences a stress level of $0.97 \le \sigma_{eqv} / \hat{\sigma}_{eqv} \le 1.05$ which is confined

within 5% of the average value. Yet, the present results indicate a high degree of homogeneity of the mechanical fields, that is quite appropriate to engineering testing of mechanical behavior of materials.

Averaging ε_{eqv} by means of Eqs. (2) yields $\hat{\varepsilon}_{eqv} = 0.386$. Fig. 8b reveals that ε_{eqv} too is not strictly uniform on the mid-section, but ranges between $0.73 \le \varepsilon_{\mathrm{eqv}} / \hat{\varepsilon}_{\mathrm{eqv}} \le 1.14$. Most of the area of the section experiences a strain level, of $0.88 \le \varepsilon_{\mathrm{eqv}} / \hat{\varepsilon}_{\mathrm{eqv}} \le 1.06$ which is confined within 12% of the average value.

Averaging σ_{hyd} by means of Eqs. (3) yields $\hat{\sigma}_{\mathrm{hyd}} = -0.378\,\mathrm{GPa}$. Fig. 8c reveals that the hydrostatic stress too, σ_{hyd} , is not uniform on the mid-section but ranges between $0.53 \le \sigma_{\mathrm{hyd}} / \hat{\sigma}_{\mathrm{hyd}} \le 1.85$. Most of the area of the section experiences a stress level of $0.82 \le \sigma_{\mathrm{hyd}} / \hat{\sigma}_{\mathrm{hyd}} \le 1.11$ which is confined within 18% of the average value.

Averaged Values of Stresses and Strains on the Mid-Section of the Gauge

Another important point is that of the relationship between the average values on the specimen mid-section and the prescribed constitutive law. While one would expect them to be automatically identical, the point must be verified. Fig. 9 shows the average stress-strain relation for two different hardening materials $\mathrm{E_p} = \mathrm{E}/100, \mathrm{E}/2000$, and four different gauge heights: $\mathrm{w} = 0.25, 0.5, 1.0, 2.0\,\mathrm{mm}$. Fig. 9 clearly indicates that the prescribed constitutive relations are recovered by all four gauges for the two hardening moduli. Additional information is gained by estimating the relative error that arises in the estimation of the plastic modulus. Table 1 summarizes results of linear regressions made on the results of Fig. 9. The result of the linear regression $\mathrm{E_p^*}$ is compared to the real hardening modulus $\mathrm{E_p}$. The maximum difference from the real modulus was found to be $33\,\mathrm{MPa}$ which is less than 1.6% error for $\mathrm{E_p} = \mathrm{E}/100$. The *relative* errors for the soft material with $\mathrm{E_p} = \mathrm{E}/2000$ are greater and reach 13% because of the small value of the hardening modulus. These results show that the specimen can be reliably used and data processed for experimental testing of materials.

The averaged hydrostatic stress $\hat{\sigma}_{hyd}$ vs. $\hat{\varepsilon}_{\mathrm{eqv}}$ is plotted in Fig. 10 for the above-mentioned cases. The figure reveals that the hydrostatic stresses are all negative and increase (in absolute value) as the gauge height decreases. The “hard” material reaches higher

hydrostatic stresses, which change more steeply with increasing strain $\hat{\varepsilon}_{eqv}$. It can also be observed in Figs. 9 and 10 that the smaller the gauge height, the greater the averaged total strain.

The variation of $\left|\hat{\sigma}_{\text{hyd}} / \hat{\sigma}_{\text{eqv}}\right|$ is plotted in Fig. 11. As before, four different gauge heights are examined: $w = 0.25, 0.5, 1.0$ and 2 mm. Fig. 11a is for hardening modulus of $E_p = E/100$ and Fig. 11b is for hardening material of $E_p = E/2000$. Fig. 11 reveals that for $E_p = E/100$ and all the gauge heights examined, this ratio is: $0.52 < \left|\hat{\sigma}_{\text{hyd}} / \hat{\sigma}_{\text{eqv}}\right| < 0.25$. For $E_p = E/2000$ and all the gauge heights examined, this ratio becomes: $0.54 < \left|\hat{\sigma}_{hyd} / \hat{\sigma}_{eqv}\right| < 0.20$. The ratio of $\left|\hat{\sigma}_{\text{hyd}} / \hat{\sigma}_{\text{eqv}}\right|$ does not change significantly during the loading of a specimen; for example: for the specific gauge height $w = 0.5$ mm and $E_p = E/2000$ this ratio is $0.39 < \left|\hat{\sigma}_{hyd} / \hat{\sigma}_{eqv}\right| < 0.48$. All the above results indicate a variable state of triaxiality, as compared to a uniaxial type of specimen. They also show that the hydrostatic to equivalent stress ratio is relatively insensitive to the constitutive behavior of the material.

Effect of Gauge Root Radius

As with any other specimen, the influence of local stress concentrations is assessed to estimate the reliability of the future experimental results obtained using this specimen. For this reason, three specimens of $w = 2$ mm and hardening modulus of $E_p = E/500$, which differ by their gauge root radius were analyzed (all other material and geometrical properties remained identical). The selected root radii are: $R_1 = 0.125, 0.25, 0.4$ mm. The distribution of the stresses $\hat{\sigma}_{eqv}$ and $\hat{\sigma}_{hyd}$ on the cross section shown in Fig. 4, are plotted as contour maps in Fig. 12. The maps were generated for the smallest root radius of $R_1 = 0.125$ mm and for applied displacement of $3\text{x}10^{-5}$ m. At this displacement level, the material adjacent to the root radius just starts to yield. It can be noticed that the effect of the corner radius is mostly confined to the corner.

The effect of the selected root radii on the $\hat{\sigma}_{\text{eqv}} - \hat{\varepsilon}_{\text{eqv}}$ graph is shown in Fig. 13. It can be observed that the three corner point radiuses yield the same characteristic mechanical behavior. Decreasing the root radius causes a slight increase in $\hat{\sigma}_{\text{eqv}}$ and $\hat{\varepsilon}_{\text{eqv}}$. The presence of

a sharp corner does not affect the ability to reconstruct the characteristic behavior of the material from the $\hat{\sigma}_{eqv} - \hat{\varepsilon}_{eqv}$ graph.

The effect of the selected root radii on the $\hat{\sigma}_{hyd} - \hat{\varepsilon}_{eqv}$ graph is shown in Fig. 14. It can be observed that decreasing the root radius causes a slight decrease in $\left|\hat{\sigma}_{hyd}\right|$.

These results indicate that when $0.125 \text{ mm} < R_1 < 0.4 \text{ mm}$, stress concentration near the root radius has a negligible influence on the calculated averaged mechanical properties.

Approximation of $\hat{\sigma}_{eqv} - \hat{\varepsilon}_{eqv}$ Curve from P - d Data

In this section, we present correlations between $\hat{\sigma}_{eqv}$ and $\hat{\varepsilon}_{eqv}$ and the applied load P and displacement d respectively. For each kind of specimen (the difference is only the gauge height) three numerical analyses are performed for three different ratios of E_p / E. For every ratio, $\hat{\sigma}_{eqv}$ is approximated by a first order polynomial function of the applied force

$$\hat{\sigma}_{eqv} \cong f_1(P) = c_1 \sigma_Y + c_2 \frac{P}{Dt} \tag{4}$$

Likewise, $\hat{\varepsilon}_{eqv}$ approximated by a second order polynomial of the form

$$\hat{\varepsilon}_{eqv} \cong f_2(d) = c_3 + c_4\left(\frac{d}{h}\right) + c_5\left(\frac{d}{h}\right)^2 \tag{5}$$

where $h = \sqrt{2}w$ is the horizontal height of the gauge, t its thickness and D is the diameter of the specimen. The coefficients $c_i^{(j)}, i = 1\cdots5, j = 1\cdots3$, where the superscript $^{(j)}, j = 1\cdots3$ corresponds to a ratio E_p / E are calculated from the numerical results by using least squares approximations with the aid of the IMSL[11]. To make Eqs (4)-(5) suitable for *any* ratio of E_p / E the coefficients are approximated by

$$c_i = a_{ij} x_j \tag{6}$$

where

$$x_j = \begin{Bmatrix} 1 \\ \ln(E_P / E) \\ \ln^2(E_P / E) \end{Bmatrix} \tag{7}$$

The coefficients a_{ij} are calculated by solving i systems of j equations. Eqs (4-6) can be used to approximate the stress-strain curve for any ratio of E_p / E. Verification problems are presented in the following section.

A demonstration of the approximations (4)-(6) is seen in Figs 15 and 16. For this demonstration, the numerical results of a specimen of $w = 2$ mm with three different plastic moduli: $E_p = E/2000$, $E_p = E/500$ and $E_p = E/100$ are used. On these figures the numerical results are shown as points, and the numerical approximation, Eqs (4)-(5), are shown as solid lines. It can be observed that a very good approximation is achieved. Table 2 summarizes the coefficients $c_i^{(j)}$ for the specimen of $w = 2.0$ mm.

Processing of Results

A First Method for the Determination of the Plastic Modulus Ep (Method #1)

The procedure to determine the plastic modulus E_p for large strains from experimental P-d data is as follows:

1. Conduct an experiment with the SCS specimen and obtain the P-d data.
2. Perform numerical analyses of the SCS specimen with three different ratios of E_p / E.
3. From the results of step 2, calculate the coefficients a_{ij} using Eqs (3)-(7).
4. Guess an initial value for E_p / E.
5. Calculate c_i, $i = 1 \cdots 5$ of Eqs (6).
6. Use Eqs (4)-(5) with the coefficients obtained in step 5 together with the experimental P-d data to obtain the $\hat{\varepsilon}_{eqv} - \hat{\sigma}_{eqv}$ graph.
7. Calculate the slope E_p^* from the $\hat{\varepsilon}_{eqv} - \hat{\sigma}_{eqv}$ graph .
8. Conduct a convergence test: if E_p^* is not equal to the assumed E_p in step 4, then guess another value for E_p / E and go to step 5. if E_p^* is equal to the assumed E_p in step 4, then terminate.

The procedure described in steps 1-8 is incorporated into a minimization procedure of the norm $\left| E_p^* - E_P \right|$.

A Second Method for the Determination of the Plastic Modulus Ep (Method #2)

For each specimen geometry, three different P-d graphs are plotted. The three graphs correspond, as mentioned, to the three different ratios of E_p / E. Each graph can be approximated in the range $d_{min} < d < d_{max}$ by a straight line. The values of d_{min} and d_{max} can be chosen arbitrarily as long as they are greater than d_Y which is the value of the applied displacement that causes the gauge of the specimen to deform plastically, from a macroscopic point of view. The slope of a segment of the P-d graph in the range $d_{min} < d < d_{max}$ is calculated by using least squares approximation with the aid of the IMSL[11]. The plastic modulus as a function of the slope is approximated by a second order polynomial

$$E_p(s) = d_0 + d_1 s + d_2 s^2 \tag{8}$$

The coefficients $d_i, i = 0 \cdots 2$ are calculated by solving the system of equations

$$E_P^{(k)} = d_0 + d_1 s^{(k)} + d_2 s^{2(k)}, \quad k = 1 \cdots 3 \tag{9}$$

The procedure to determine the plastic modulus E_p for large strains is thus:

1. Perform an experiment with the SCS specimen and obtain the P-d data.
2. Perform three numerical analyses of the SCS specimen with three different ratios of E_p / E.
3. Plot the three different P-d numerical graphs of step 2 with the experimental P-d graph obtained in step 1.
4. Select the values d_{min} and d_{max}.
5. Calculate the three numerical slopes $s^{(k)}, k = 1 \cdots 3$ in the range $d_{min} < d < d_{max}$.
6. Calculate The coefficients $d_i, i = 0 \cdots 2$ according to equation (9).
7. Calculate the slope s^* in the range $d_{min} < d < d_{max}$ from the experimental P-d graph.
8. Obtain an approximation for E_P^* by substituting s^* in equation (8).

The procedure described in steps 1-8 is incorporated into a computer code which finds automatically the value of E_p^*.

A Simple Method to Determine the Stress-Strain Relationship

The stress-strain curve determination was presented originally as a simple expression[7] of three coefficients, k_i （i=1-3）, as follows:

$$\hat{\sigma}_{eqv} = k_1\left(1 - k_2\hat{\varepsilon}_{eqv}\right)\frac{P}{Dt} \tag{10}$$

$$\hat{\varepsilon}_{eqv} = k_3\frac{d}{h} \tag{11}$$

One would naturally like to develop similar relations that are quite easy to use for the practitioner. Therefore, once E_p^* has been determined using either approach described earlier, its value is substituted into Eqs (6) to calculate $c_i, i = 1\cdots5$. The experimental load-displacement data along with Eqs (4)-(5) are used to plot the desired stress-strain curve. The curve is fitted with the aid of least squares to determine the three coefficients k_1, k_2 and k_3. These coefficients, once determined for a given material and gauge geometry, will be used to process all the subsequent experimental data.

The Dependence of the k_i Coefficients on Material Properties and Gauge Geometry

Specimens with two different gauge heights were considered: $w = 1$ mm and $w = 2$ mm. The material properties which were used in all analyses were: $E = 73.8$ MPa , $\nu = 0.33$ and $\sigma_Y = 400$ MPa. A bi-linear material model was assumed. Three different plastic modulus were analyzed for each gauge height: $E/E_p = 100, 500$ and 2500 . The k_1 , k_2 and k_3 coefficients were calculated with the aid of least squares for each case as shown in Table 3. From Table 3 it can be concluded that the coefficient k_1 depends on the gauge height and almost not on the material properties. On the other hand, k_2 and k_3 are dependent on both the gauge height and material properties.

Verification Problems

In order to validate the two data reduction methods and the accuracy of Eqs. (4)-(5) for E_p/E other than those investigated, two additional cases were solved numerically. The first case involves a specimen with $w = 2$ mm and $E_p/E = 1/300$. The second case involves a specimen with $w = 0.25$ mm and $E_p/E = 1/800$. The obtained numerical (P,d) data served as the "experimental measured results" input for the two processing methods.

For the first case, applying method #1 in the range $0.1\ \text{mm} \le d \le 1\ \text{mm}$ yields $E_p^*/E = 1/311$. Applying method #2 to the same d range yields $E_p^*/E = 1/300$. The coefficients c_i, $i = 1...5$, are obtained by substitution of the average value $\hat{E}_p^*/E = 1/306$ into Eqs 6. Eqs (4)-(5) for the first case become

$$\hat{\sigma}_{eqv} = 0.2837 \cdot 700 \cdot 10^6 - 0.6850\frac{P}{Dt} \tag{12}$$

$$\hat{\varepsilon}_{eqv} = -0.0090 + 1.0663\left(\frac{d}{h}\right) - 0.4258\left(\frac{d}{h}\right)^2 \tag{13}$$

For the second case, applying method #1 to the range $0.2\ \text{mm} \le d \le 0.8\ \text{mm}$ yields $E_p^*/E = 1/828$. Applying method #2 to the same d range yields $E_p^*/E = 1/807$. The coefficients c_i, $i = 1...5$, are obtained by substitution of the average value $E_p^*/E = 1/817$ into Eqs 6. Eqs (4)-(5) for the second case become

$$\hat{\sigma}_{eqv} = 0.3042 \cdot 700 \cdot 10^6 - 0.5911\frac{P}{Dt} \tag{14}$$

$$\hat{\varepsilon}_{eqv} = -0.01 + 0.3033\left(\frac{d}{h}\right) - 0.0191\left(\frac{d}{h}\right)^2 \tag{15}$$

From Eqn. (4) it is clear that when $\hat{\sigma}_{eqv} = \sigma_Y$, $P = P_Y$. For this reason in Eqs (12) and (14) the coefficient c_1 is slightly modified to

$$c_1^* = 1 - \frac{c_2}{\sigma_Y}\frac{P_Y}{Dt} \tag{16}$$

Pairs of data (P,d) which were calculated by the numerical analyses were substituted into Eqs. (12)-(15) and the resulting curves of $\hat{\varepsilon}_{eqv} - \hat{\sigma}_{eqv}$ are plotted in Fig. 17. The "real" bi-linear $\hat{\varepsilon}_{eqv} - \hat{\sigma}_{eqv}$ curves which were used in the numerical analyses are compared to the approximation by (12)-(15). Since there is a very good agreement in Fig. 17 for the two cases corresponding to the two extreme gauge heights, it can be concluded that the processing methods as well as the approximations are suitable for assessing the constitutive relations, noting that the higher gauge yields more accurate results.

Experimental Results

SCS specimens made of tungsten heavy alloy were tested for obtaining their characteristic $\hat{\varepsilon}_{eqv} - \hat{\sigma}_{eqv}$ behavior. The results were processed as described in the previous sections. Two parallelepiped SCS specimens with the three different gauge widths: 0.3 mm, 1 mm and 2.5 mm, were tested. The specimens were typically 20 mm high, 9.5 mm wide, and 6.5 mm thick. The gauge thickness was 2.5 mm. The calculated $k_i, i = 1...3$ for each gauge width are summarized in Table 4. The SCS results are compared to results obtained by compressing cylinders. The latter are limited to small strains because of barreling, and three typical results are plotted in Fig.18. For this material, $E/E_p \approx 600$. A good agreement is observed for the two specimen geometries in the comparable range of small strains. It can also be noted that, to a first approximation, this material behaves in a bi-linear fashion. Consequently, the results are believed to be reliable at larger strains for the SCS.

Discussion

- The Shear Compression Specimen has been modeled and thoroughly analyzed using the finite element method. Several previously unaddressed issues have been investigated and reported: the influence of specimen geometry and material parameters. In addition, a simple data reduction technique has been developed and validated (Eqs. 10 and 11). The main points of this study will now be discussed and summarized. The numerical calculations assume a bi-linear elastic-plastic material. While such a behavior is observed in some materials, the more general constitutive response is of the parabolic type. Consequently, the present assumption is an approximation which should be carefully checked for each specific type of material. However, it is felt that as a first approximation, it is satisfactory, keeping in mind that a parametric study of the present time would be difficult to achieve for a parabolic material.
- The analyses reveal that the averaged $\hat{\varepsilon}_{eqv} - \hat{\sigma}_{eqv}$ on the mid-section reproduces the constitutive bi-linear equation of the material. This result is not influenced by the yield stresses or specimen geometry (gauge height, root radius). This result is backed by the observation that the stress and strain distribution in the gauge section of the specimen is reasonably homogeneous. Previous work[7] had not addressed this important issue.

- Stress concentration in the gauge fillet has been examined in detail for three root radii that are representative of most specimens. Calculations were carried out for quasi-static loading only, but one can reasonably assume that the main findings apply to the dynamic case as well. The results indeed show stress concentration, as expected, however it does not affect significantly the determination of the stress-strain relationship. One should nevertheless keep in mind that specific issues related to stress concentration may indeed be affected. As an example, shear localization (such as adiabatic shear banding), may indeed be triggered by the local mild stress concentration observed. However, this point is beyond the scope of the present work, as it essentially requires additional experimental and numerical work.
- The simple relationship originally proposed by Rittel et al.[7], as expressed in Eqs. (10) and (11), was further verified in this work, and can now be simply established for any material and gauge geometry. The applicability of the quasi-static procedure to the dynamic case will be verified in part II of this work. The k_i coefficients are related to some extent to the actual plastic modulus. The degree of influence is related to the plastic modulus itself, mostly for k_2 and k_3. However, it was found that these coefficients are almost unaffected by the material properties and gauge width, when the latter is wider (Table 3). Therefore, for a given material, the best accuracy is expected when testing "large" gauge specimens (e.g., w=2mm). The average strain is dictated by k_3, so that prior knowledge on the anticipated plastic properties will be useful in planning the experiments for an anticipated strain. Keeping in mind the above-mentioned, this relationship is of huge practical interest, and is the first of its kind for such a shear specimen.

Finally, the present work confirms that the SCS geometry can confidently be used for the characterization of large strain behavior of materials. As such, the present results are among the most thorough analyses presented for new specimens of that kind.

Conclusions

The SCS has been thoroughly analyzed numerically for quasi-static loading conditions. Bi-linear material behavior has been assumed, as a first step for the present analysis. Specific issues that were not addressed previously have been investigated, and the following conclusions can be drawn from the present work:

- The stresses and strains are quite homogeneous in the gauge section.
- Stress concentration at the gauge fillets indeed exists but is of minor influence on the determined mechanical properties.
- Very simple relations have been developed to reduce the load-displacement curve into an equivalent stress-strain relationship in terms of 3 coefficients. These coefficients are material and gauge geometry dependent, and guidelines have been provided for their determination.
- Additional work should be carried out to include other types of material behavior, such as parabolic.

Acknowledgement

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TABLES

Table 1
Comparison between computed hardening modulus by linear regression and the real hardening modulus

$w \quad [mm]$ gauge height	$E_p \quad [10^9 Pa]$	$E_p^* \quad [10^9 Pa]$ linear regression	$E_p - E_p^* \quad [10^9 Pa]$ difference	$(E_p - E_p^*)/E_p$ difference [%]
0.25	2.1	2.102	-0.002	-0.1
0.50	2.1	2.133	-0.033	-1.6
1.0	2.1	2.127	-0.027	-1.3
2.0	2.1	2.120	-0.020	-1.0
0.25	0.105	0.103	0.002	1.9
0.50	0.105	0.100	0.006	5.3
1.0	0.105	0.091	0.014	13.2
2.0	0.105	0.102	0.003	2.9

Table 2

Coefficients of least squares approximation for a specimen with $w = 2.0\ mm$. Note that c_1 is divided by the assumed yield stress of 700MPa.

E_P / E	$c_1 / 700 \times 10^6$	c_2	c_3	c_4	c_5
1/2000	0.6842	-0.2835	-0.0139	1.1895	-0.3575
1/500	0.3527	-0.5982	-0.0109	1.1216	-0.4186
1/100	0.1037	-0.8329	-0.0029	0.8806	-0.4136

Table 3

Typical dependence of the coefficients k_1, k_2 and k_3 on the specimen's gauge height and plastic modulus.

$w\ [mm]$	E/E_p	k_1	k_2	k_3
1	100	0.856	0.056	0.501
	500	0.851	0.142	0.649
	2500	0.848	0.174	0.691
2	100	0.928	0.212	0.739
	500	0.924	0.221	0.915
	2500	0.922	0.231	0.988

Table 4

The calculated coefficients k_1, k_2 and k_3 for analyzing tungsten heavy alloy.

w	k_1	k_2	k_3
0.3 mm	0.861	0.152	0.213
1.0 mm	0.927	0.219	0.628
2.5 mm	0.955	0.276	0.931

CAPTIONS OF FIGURES

Fig. 1 The shear compression specimen dimensions**.**

Fig. 2 The meshed half SCS.

Fig. 3 Constrained surfaces.

Fig. 4 The inclined mid-section of the gauge where the averaged $\hat{\sigma}_{eqv}$, $\hat{\sigma}_{hyd}$ and $\hat{\varepsilon}_{eqv}$ are calculated.

Fig. 5: Resultant load-displacement curves for SCS specimens.

Fig. 6 Resultant load-displacement plots for $\sigma_Y^{(1)} = 700$ MPa and $\sigma_Y^{(2)} = 1200$ MPa.

Fig. 7 Plastic zones for a specimen of $w = 2$ mm with enforced displacement of $d = 1.75$ mm. (a) Plastic zones for $E_p = E/2000$, (b) Plastic zones for $E_p = E/100$.

Fig. 8 Typical variation of σ_{eqv}, ε_{eqv} and σ_{hyd} on the section shown in Figure 4 for a 2 mm height gauge subjected to a displacement of 1.75 mm with $E_p = E/100$. (a) Contour map of σ_{eqv}. (b) Contour map of ε_{eqv}. (c) Contour map of σ_{hyd}.

Fig. 9 $\hat{\sigma}_{eqv}$ vs. $\hat{\varepsilon}_{eqv}$ for specimens of gauge heights 0.25, 0.5, 1, and 2 mm made of two different materials which differ in their hardening modulus: E/100 and E/2000. Note that the prescribed constitutive law is exactly represented by these average values.

Fig. 10 Averaged hydrostatic stresses vs. the averaged equivalent total strain on the mid section of Fig. 4.

Fig. 11 The variation of the ratio between $\hat{\sigma}_{hyd}$ to $\hat{\sigma}_{eqv}$ on the mid section vs. $\hat{\varepsilon}_{eqv}$. (a) $E_p = E/100$. (b) $E_p = E/2000$.

Fig. 12 Contour maps of $\hat{\sigma}_{eqv}$ and $\hat{\sigma}_{hyd}$ on the cross section A-A of figure 1. (a) $\hat{\sigma}_{eqv}$ (b) $\hat{\sigma}_{hyd}$

Fig. 13 $\hat{\sigma}_{eqv} - \hat{\varepsilon}_{eqv}$ graph of a $w = 2$ mm gauge with hardening modulus of $E_p = E/500$ made with three different corner point radiuses: $R_1 = 0.125, 0.25, 0.4$ mm.

Fig. 14 $\hat{\sigma}_{hyd} - \hat{\varepsilon}_{eqv}$ graph of a $w = 2$ mm gauge with hardening modulus of $E_p = E/500$ made with three different corner point radiuses $R_1 = 0.125, 0.25, 0.4$ mm.

Fig. 15 Numerical results of $\hat{\sigma}_{eqv}$ vs. P/Dt and their least squares polynomial approximation.

Fig. 16 Numerical results of $\hat{\varepsilon}_{\mathrm{eqv}}$ vs. d/h and their least squares polynomial approximation.

Fig. 17 A comparison between numerical analysis and approximations of the $\hat{\sigma}_{eqv} - \hat{\varepsilon}_{eqv}$ curve.

Fig. 18 $\hat{\sigma}_{\mathrm{eqv}} - \hat{\varepsilon}_{\mathrm{eqv}}$ curves of tungsten alloy obtained by using five SCS specimens and three cylindrical specimens.

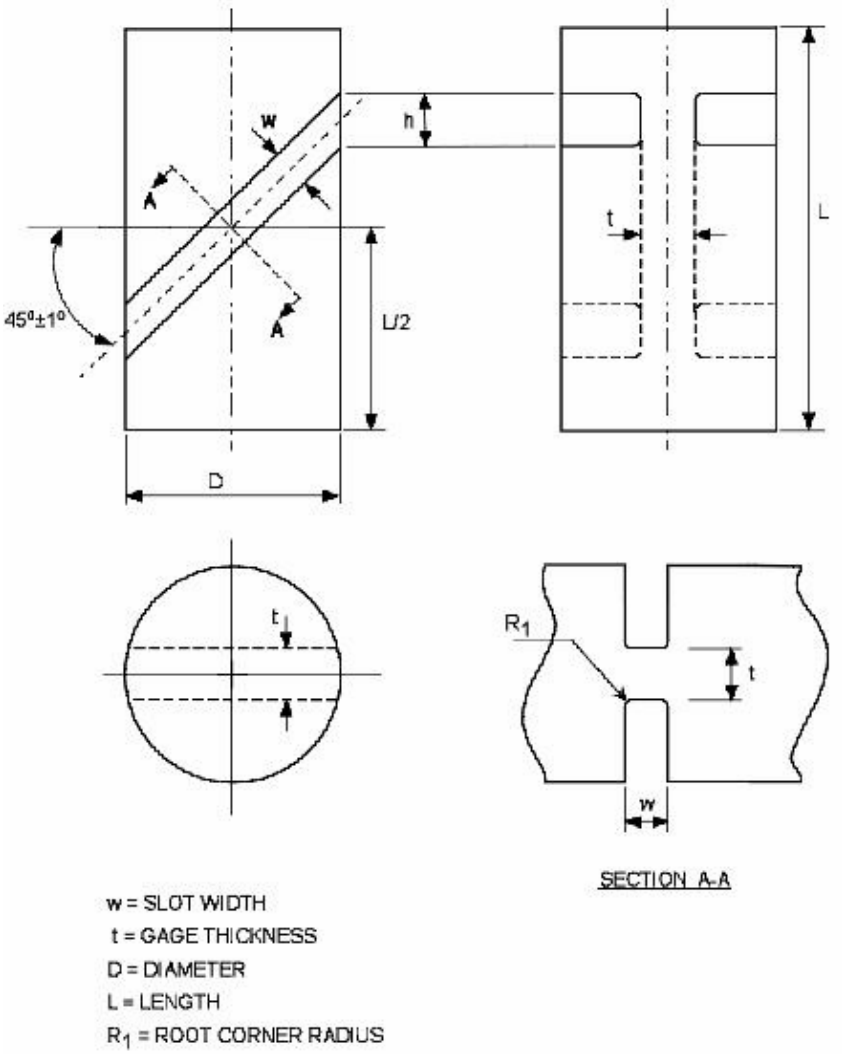


Fig. 1- The shear compression specimen.

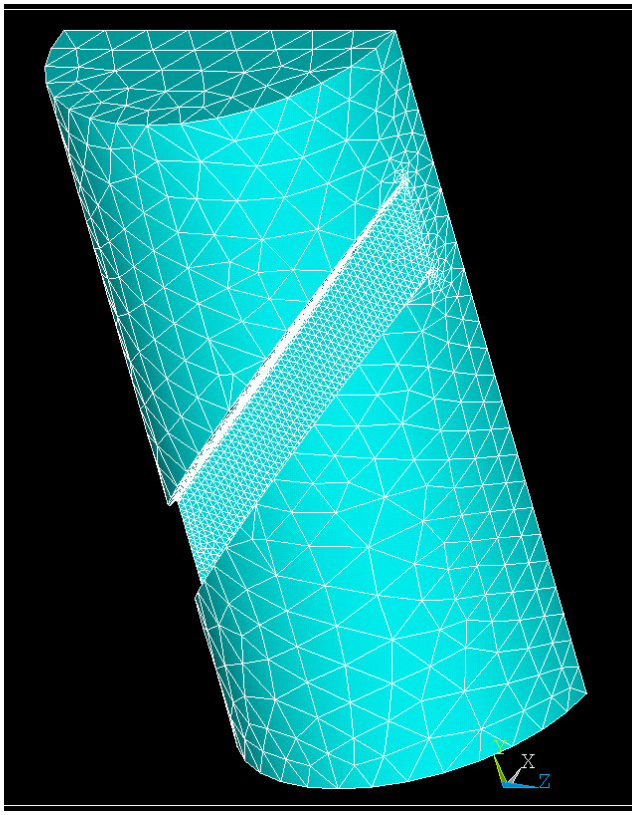


Fig. 2- The meshed half SCS.

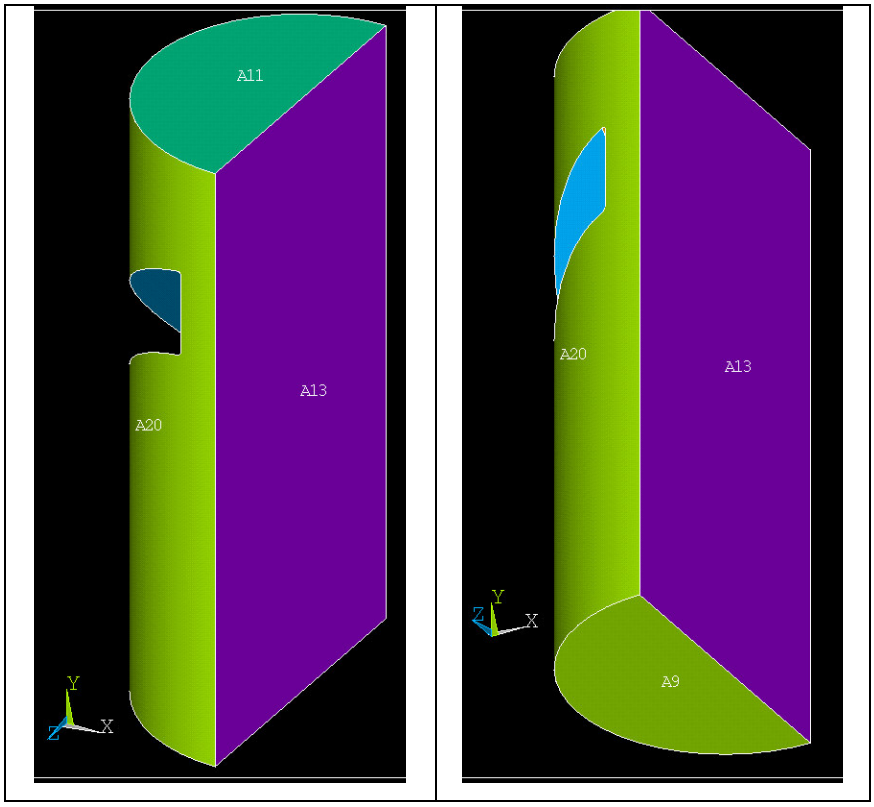


Fig. 3- Constrained surfaces.

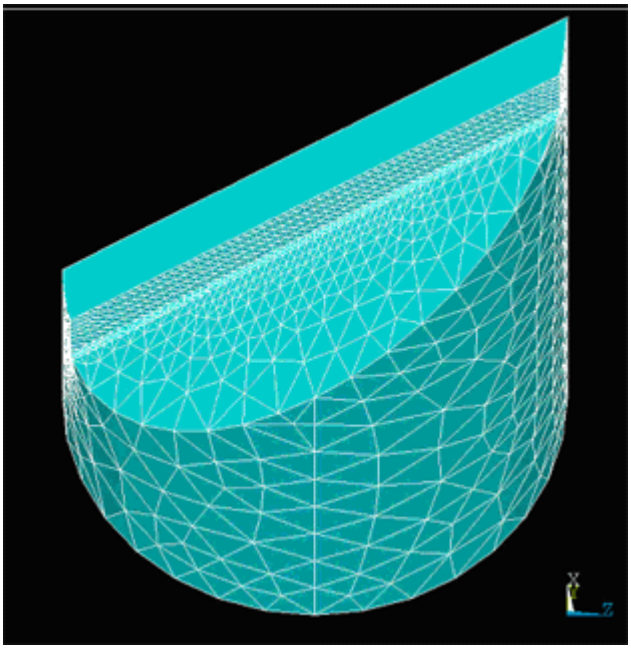


Fig. 4- The inclined mid-section of the gauge where the averaged $\hat{\sigma}_{\text{eqv}}$, $\hat{\sigma}_{\text{hyd}}$ and $\hat{\varepsilon}_{\text{eqv}}$ are calculated.

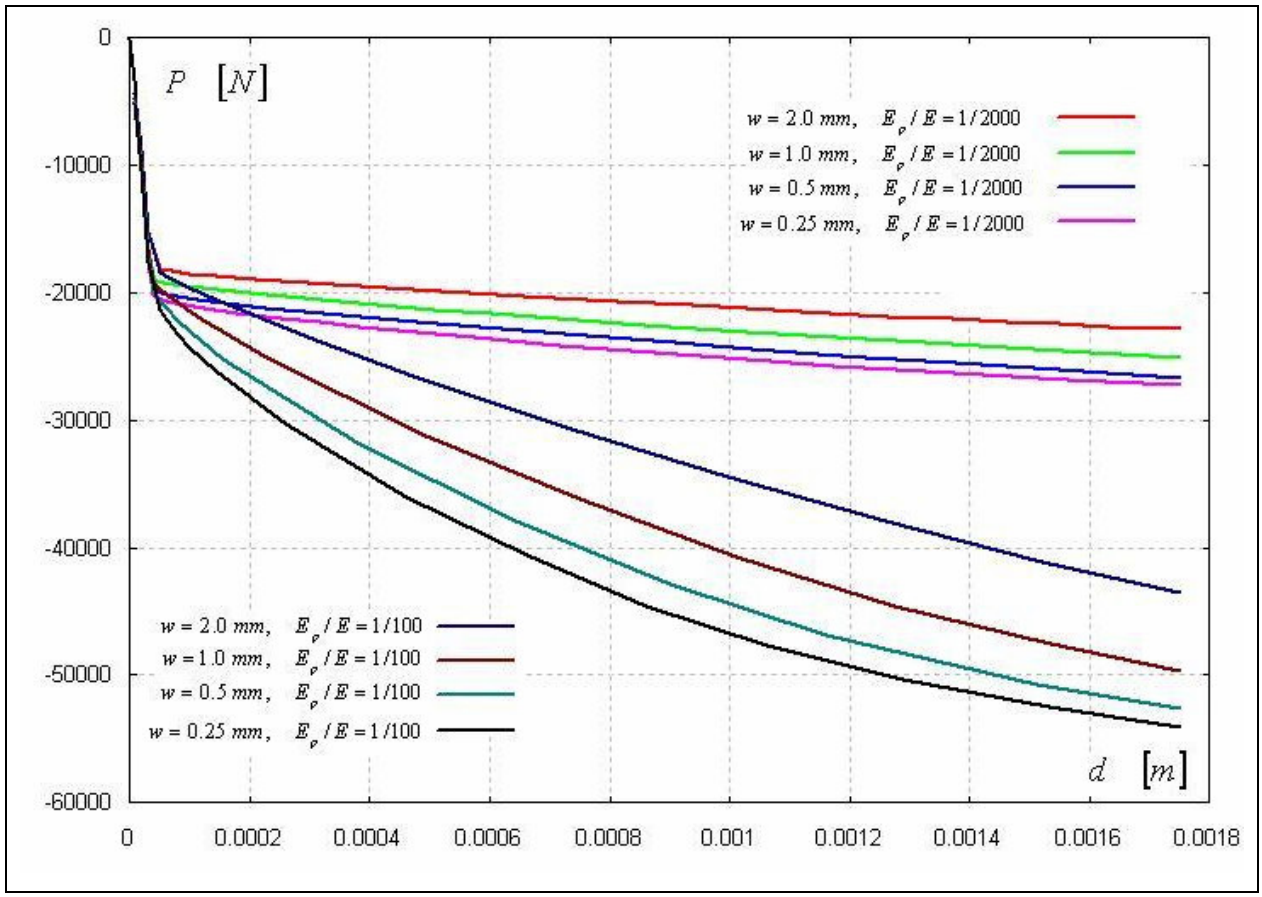


Fig. 5- Resultant load-displacement curves for SCS specimens

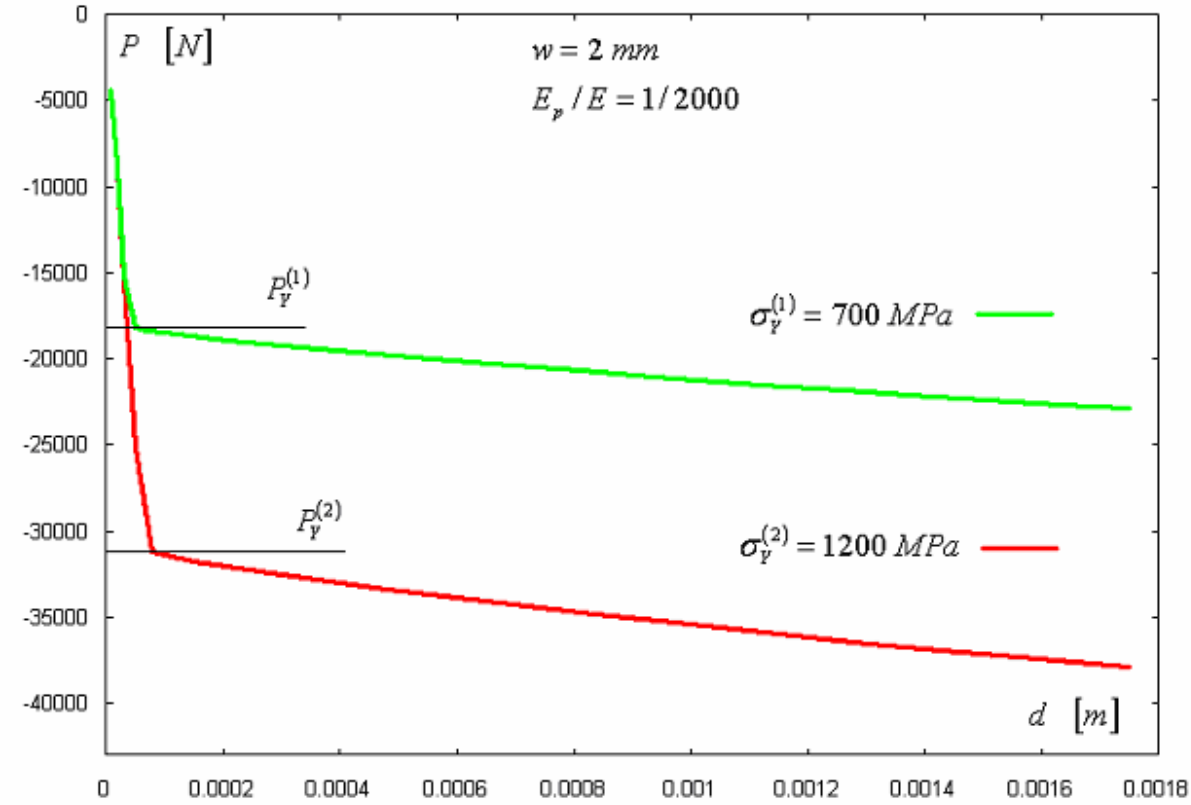
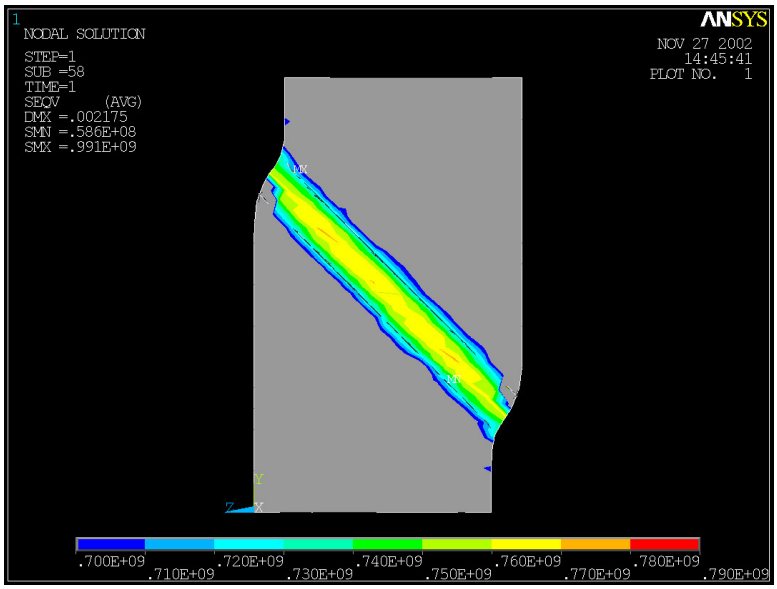
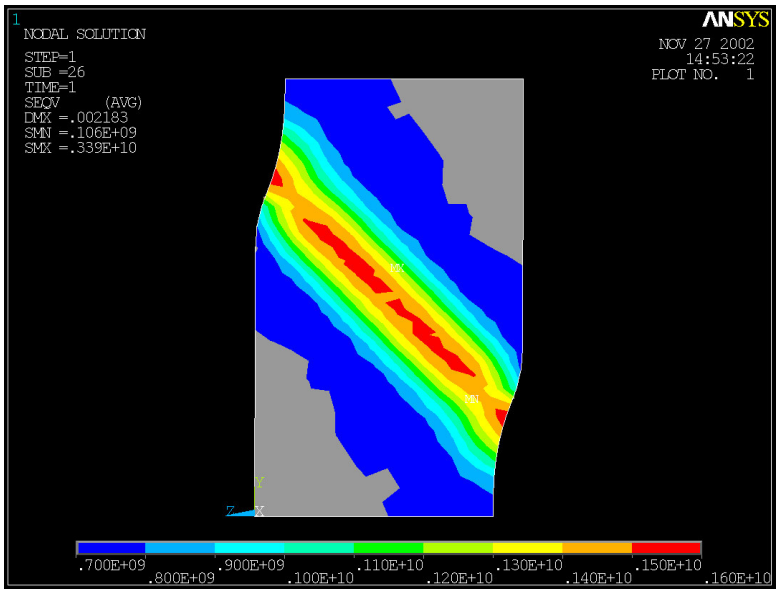


Fig. 6- Resultant load-displacement plots for $\sigma_{\mathrm{Y}}^{(1)} = 700$ MPa and $\sigma_{\mathrm{Y}}^{(2)} = 1200$ MPa .

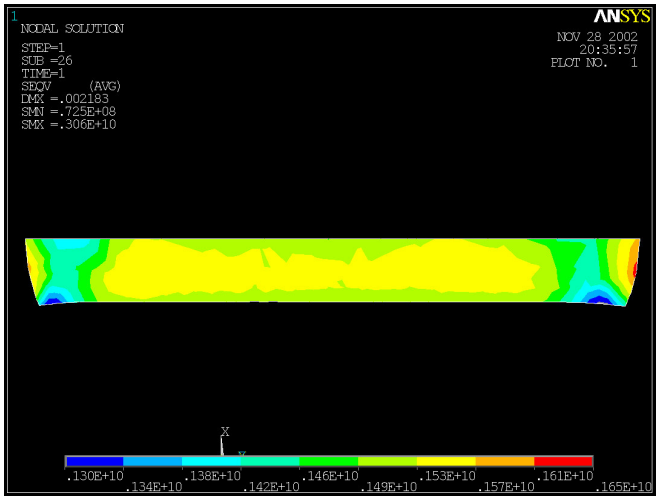


7a

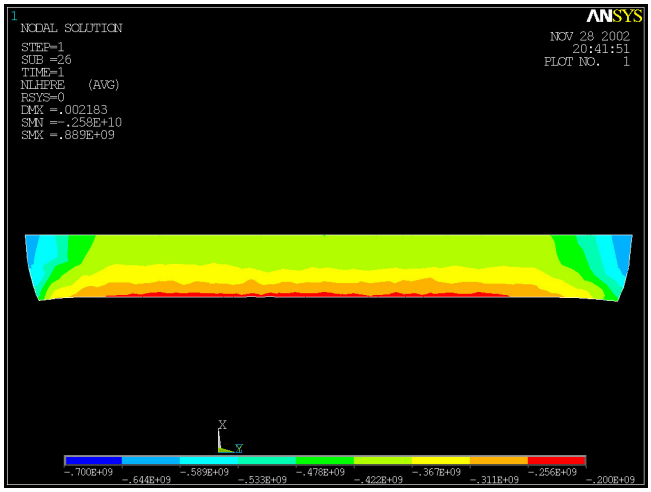


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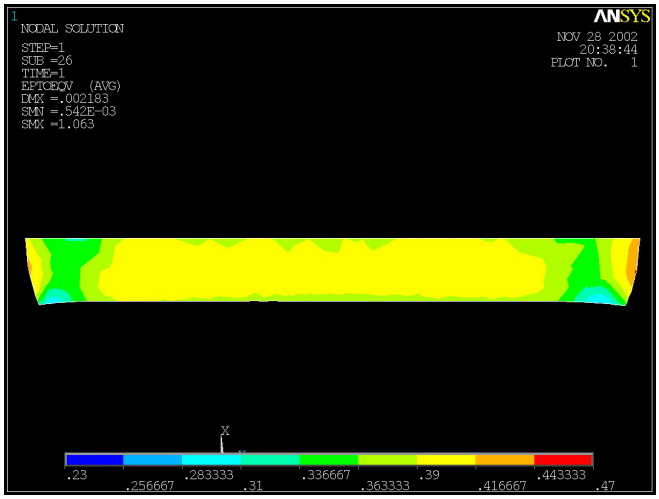
Fig. 7- Plastic zones for a specimen of $w = 2$ mm with enforced displacement of $d = 1.75$ mm. (a) Plastic zones for $E_p = E/2000$, (b) Plastic zones for $E_p = E/100$.



8a



8b



8c

Fig. 8- Typical variation of σ_{eqv}, ε_{eqv} and σ_{hyd} on the section shown in Figure 4 for a 2 mm height gauge subjected to a displacement of 1.75 mm with $E_p = E/100$. (a) Contour map of σ_{eqv}. (b) Contour map of ε_{eqv}. (c) Contour map of σ_{hyd}.

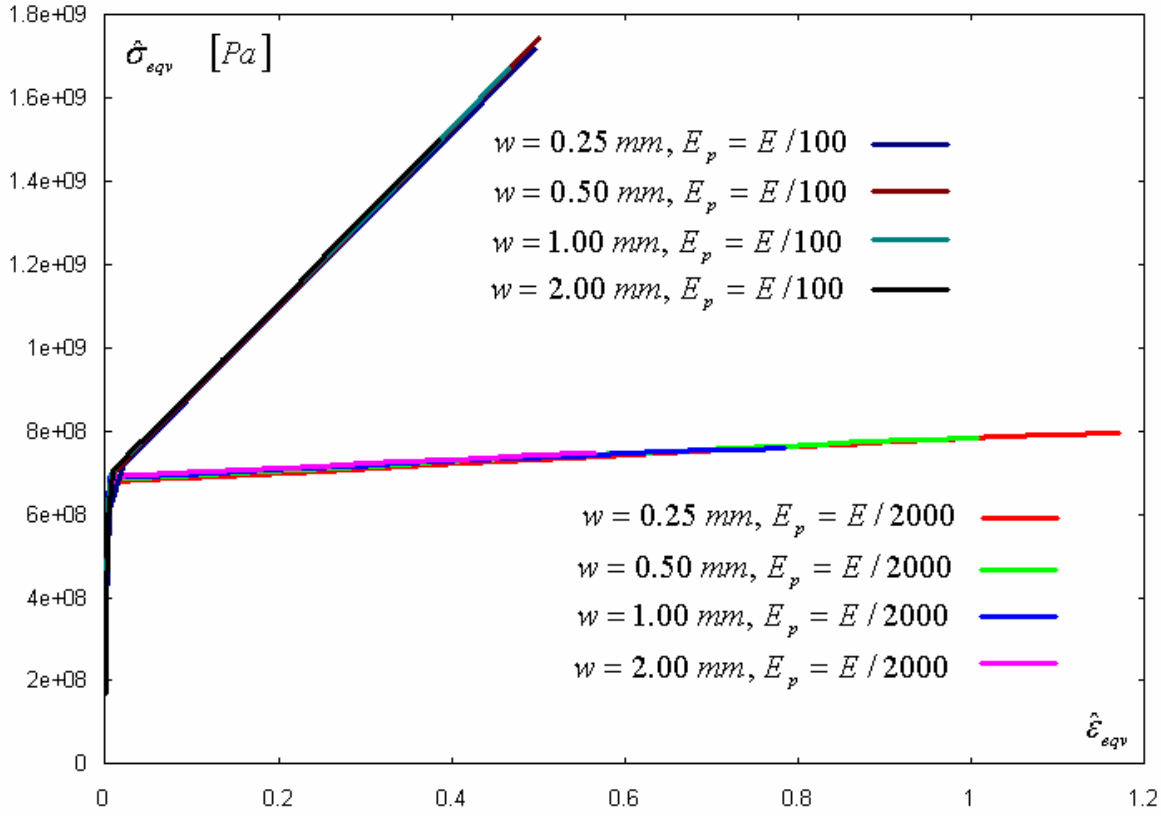


Fig. 9- $\hat{\sigma}_{eqv}$ vs. $\hat{\varepsilon}_{eqv}$ for specimens of gauge heights 0.25, 0.5, 1, and 2 mm made of two different materials which differ in their hardening modulus: E/100 and E/2000. Note that the prescribed constitutive law is exactly represented by these average values.

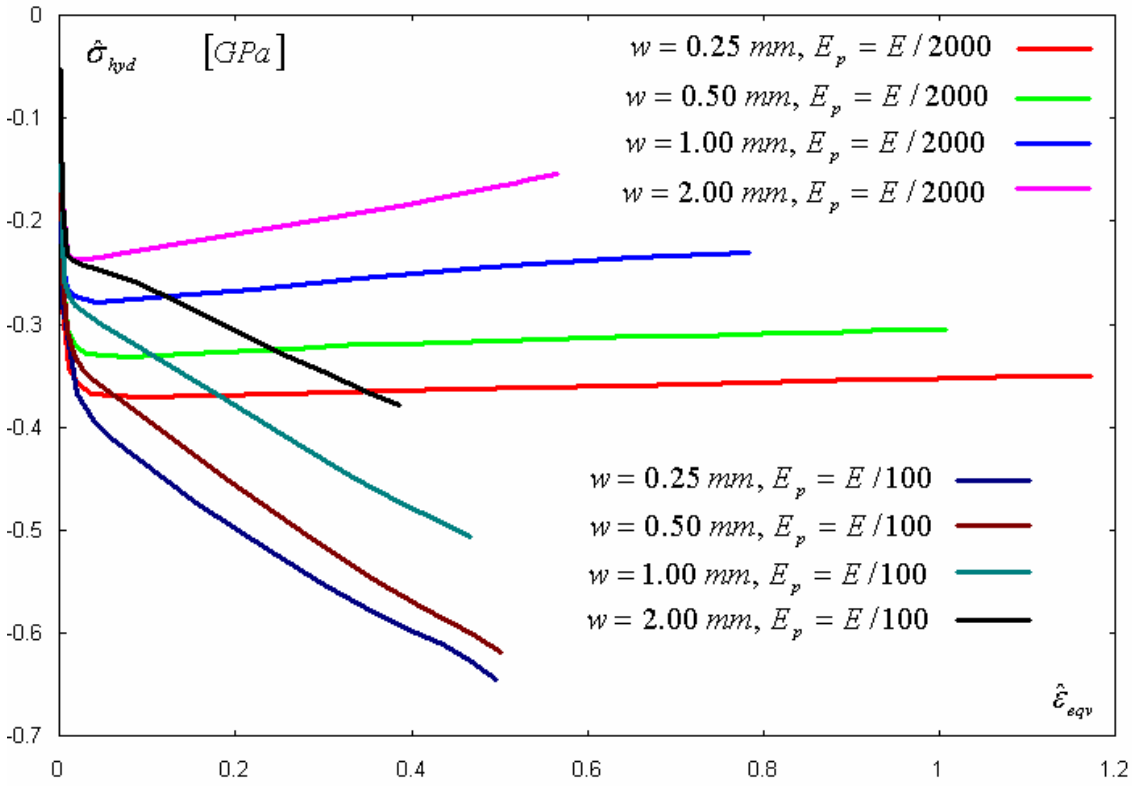
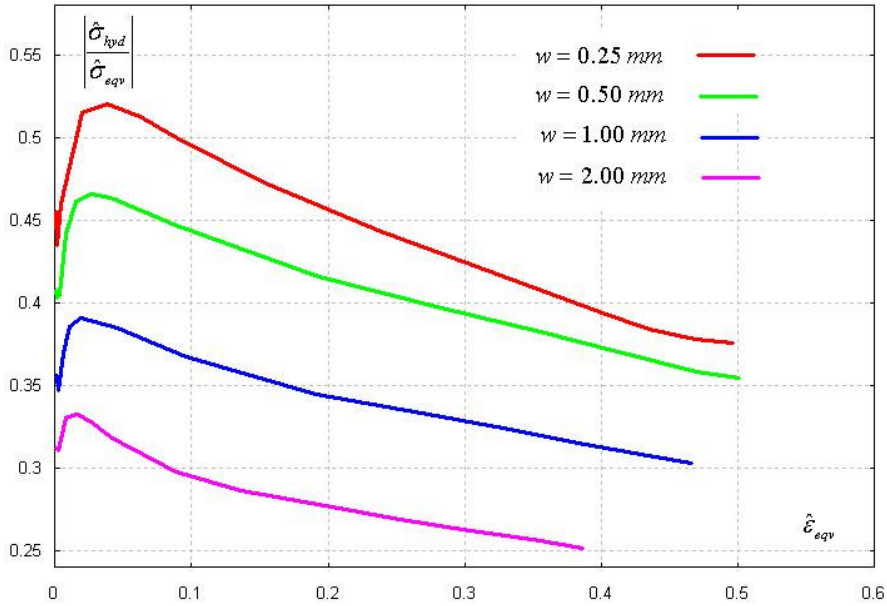
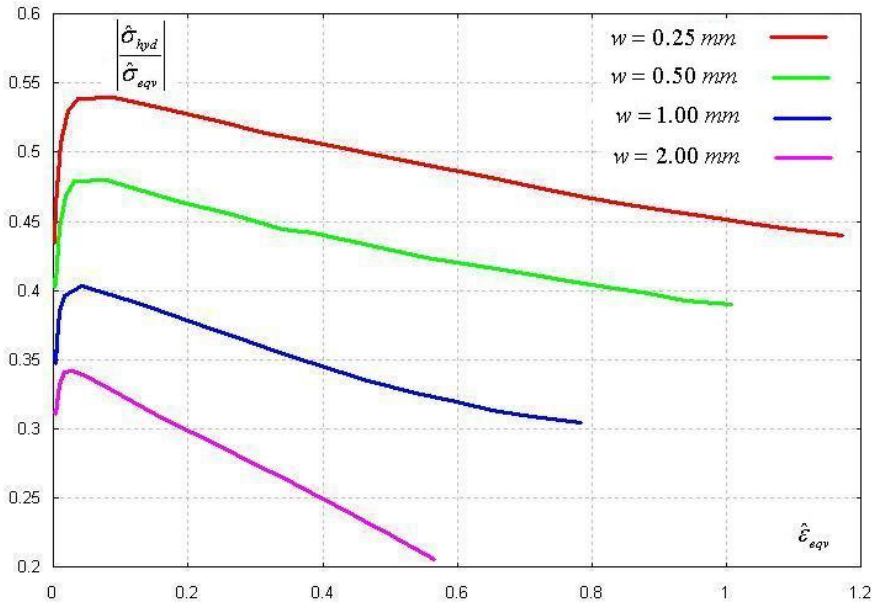


Fig. 10- Averaged hydrostatic stresses vs. the averaged equivalent total strain on the mid-section of Fig. 4.

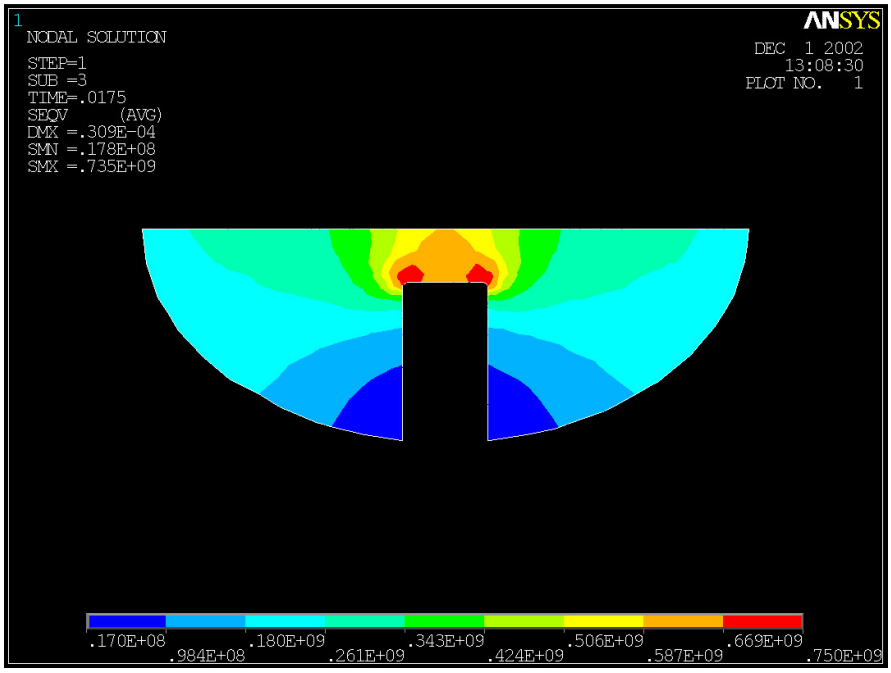


11a



11b

Fig. 11- The variation of the ratio between $\hat{\sigma}_{hyd}$ to $\hat{\sigma}_{eqv}$ on the mid section vs. $\hat{\varepsilon}_{eqv}$.
(a) $E_p = E/100$. (b) $E_p = E/2000$.



12a



12b

Fig. 12- Contour maps of $\hat{\sigma}_{eqv}$ and $\hat{\sigma}_{hyd}$ on the cross section A-A of Fig. 1. (a) $\hat{\sigma}_{eqv}$ (b) $\hat{\sigma}_{hyd}$

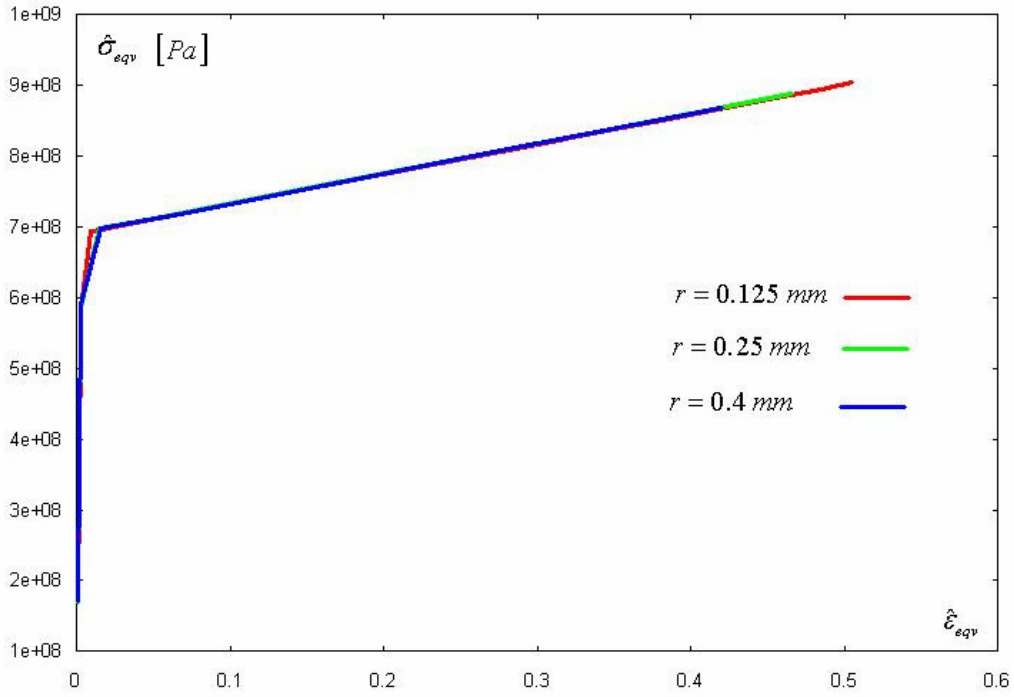


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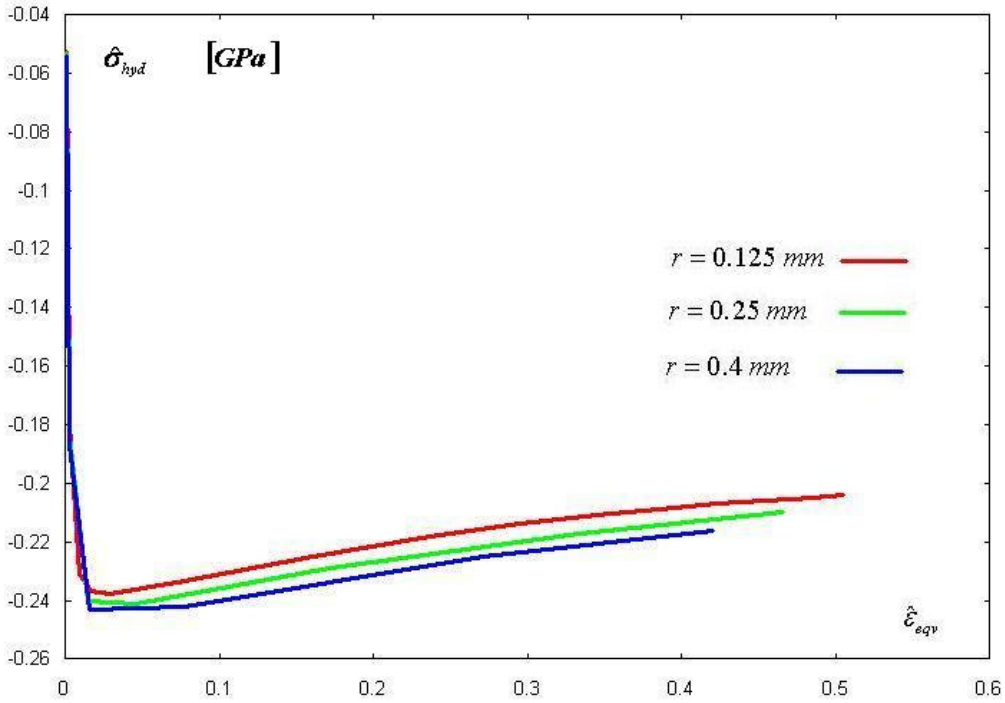


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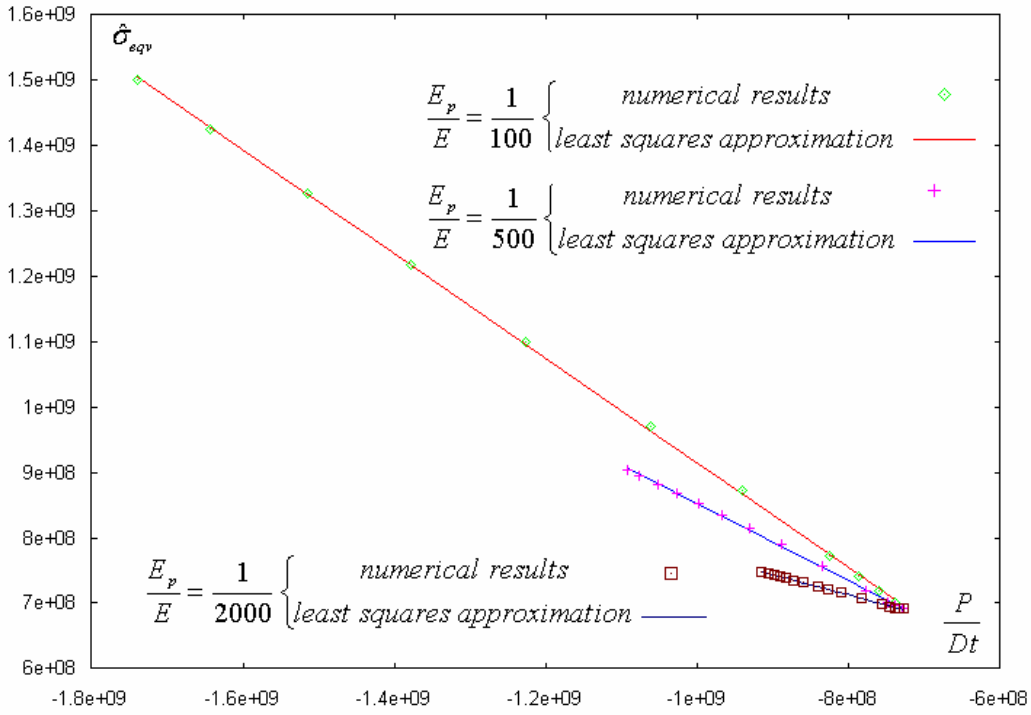


Fig. 15- Numerical results of $\hat{\sigma}_{eqv}$ vs. P/Dt and their least squares polynomial approximation.

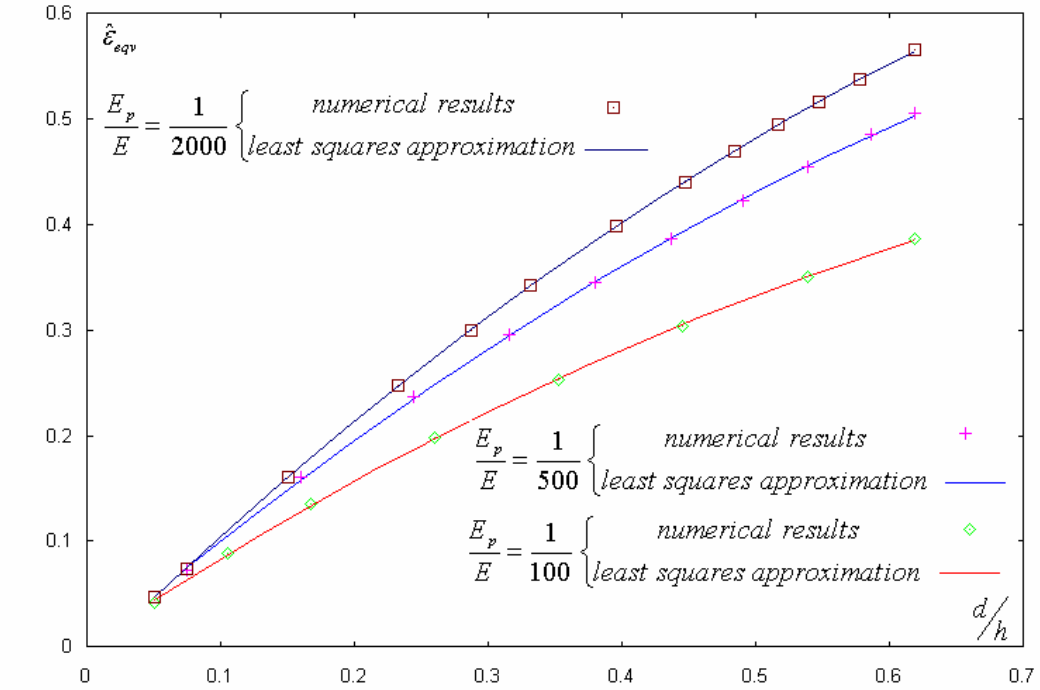


Fig. 16- Numerical results of $\hat{\varepsilon}_{eqv}$ vs. d/h and their least squares polynomial approximation.

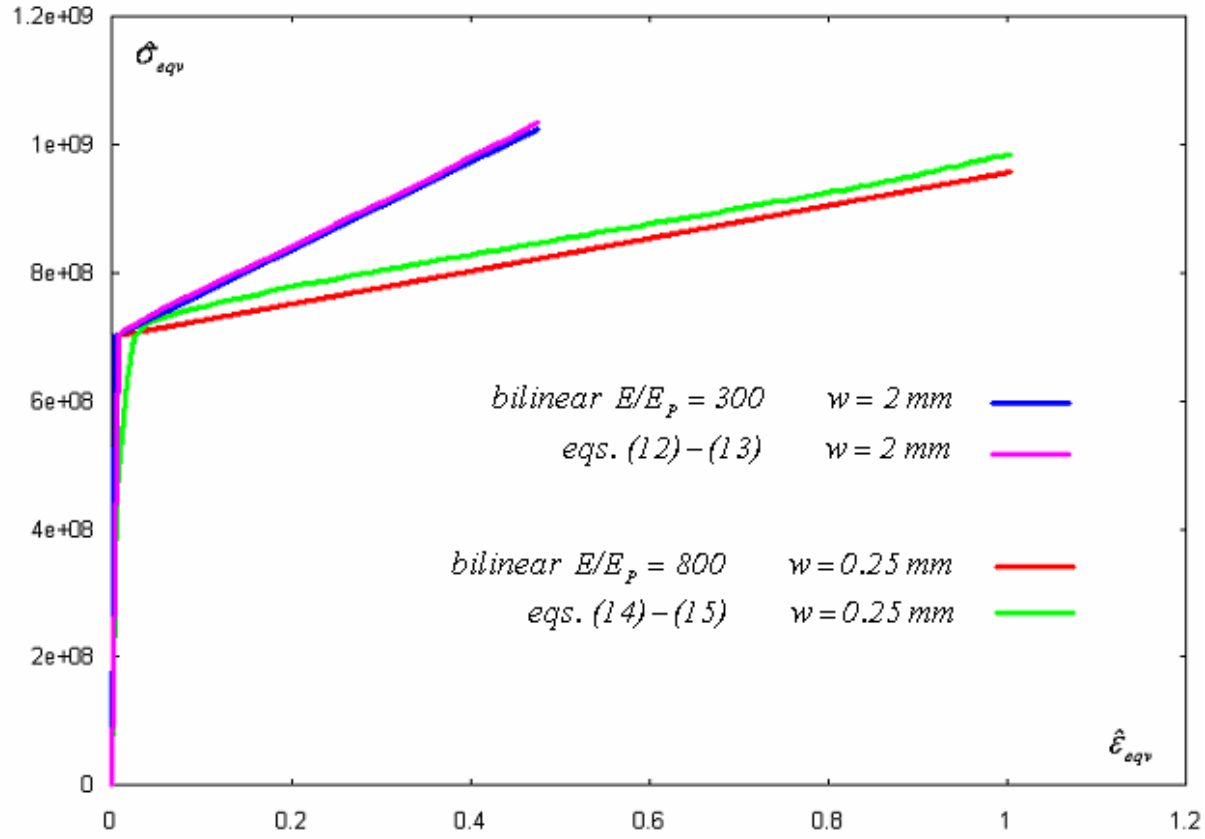


Fig. 17- A comparison between numerical analysis and approximations of the $\hat{\sigma}_{\mathrm{eqv}} - \hat{\varepsilon}_{\mathrm{eqv}}$ curve.

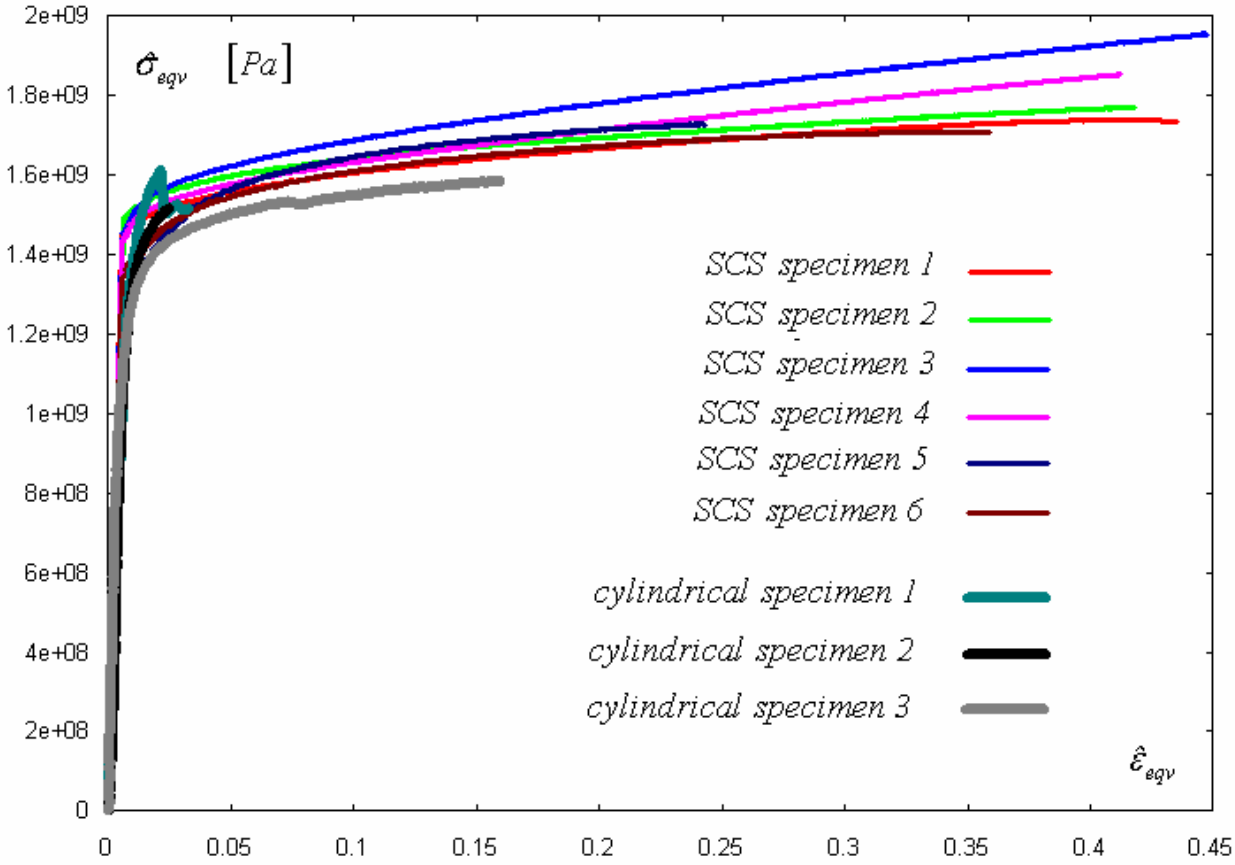


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